A Market World Model for Public LLMs: Complete, Honest, Auditable Crypto Cognition before Decision

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Abstract

Most quant stacks start from *strategy*. Public LLMs (GPT-, DeepSeek-, GLM-, and Gemini-class agents) that act in cryptocurrency markets fail earlier: they lack a complete, honest, auditable *market cognition base*. We present an anonymized *financial world-model runtime* that compiles asynchronous multi-band evidence into a point-in-time (PIT) world bundle. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{\mathrm{AI}}$; completeness, honesty (B, U, H), and WMI/ACWMI support a typed abstention contract; evidence IDs make actions auditable. Four live public LLMs (temperature 0) on the *same* 100 OOS asset-days per arm show a behavior ladder across information regimes: asked *blind* ("how should I trade BTC today?", no feed) every model refuses on 100% of days; given a raw momentum fragment the same models trade into sparse support (abstain 0.04–0.75); given the compiled world they abstain on all thin days (1.0) and act only with verifiable support; Ungated (same content, no hard flag) averages 0.68 (range 0.43–0.86). Disclosure helps but is vendor-dependent; the typed boolean enforces full cross-vendor refusal, and models that act under Ungated lose money on thin support. A non-trading grounding workflow shows the same asymmetry for analysis: with the compiled three-band bundle, models recover ready/missing sets and tilt signs near-perfectly (mean F1/acc. 0.97); from a raw feed the same scores collapse to 0.04–0.23. Economically, the compiled world state outperforms traditional baselines out-of-sample: the world-model content rule earns Sharpe 0.76 / CE +0.13 versus momentum 0.10 / −0.21 and buy-and-hold −1.40 / −1.16 (pre-specified ΔCE=0.334, p=0.034), and live LLMs consuming the bundle avoid the losses that raw-feed agents incur (mean ΔCE +1.07). We detail the runtime architecture (collectors, vintage store, BandPIT clock, bundle schema, LLM adapter) as a systems contribution for ICAIF.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics.*

Keywords

world models, public LLMs, AI agents, cryptocurrency, point-in-time, abstention, trustworthy AI, market cognition, systems

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1 Introduction

Public LLMs are increasingly asked to analyze or trade crypto markets. The bottleneck is not another alpha signal. It is *understanding*: before decision, an agent needs a market world that is **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [11, 13]. Generative world models [5, 6, 9] and tool-using finance agents [18–20] typically assume a usable observation stream. Crypto agents often do not have one.

Thesis. We build a market *world-model runtime* for public LLMs: compile raw multi-band evidence into a PIT-safe cognition bundle that GPT / DeepSeek / GLM / Gemini-class models can call. Understanding is the product objective; strategy metrics then confirm the compiled world carries real economic content—the world-model rule outperforms traditional momentum and buy-and-hold baselines out-of-sample. In slogan form: *quant from strategy; this runtime from understanding.*

What we claim. RQ1 tests whether a typed world interface can *enforce* world-conditional abstention across live public LLMs—a runtime contract, not a claim that models spontaneously discover thinness from raw ticks. RQ2 tests whether the compiled world makes analyst-facing epistemic answers verifiable. RQ3 tests whether the compiled world state delivers economic value over traditional baselines: out-of-sample, the world-model content rule outperforms same-input momentum and buy-and-hold on both Sharpe and certainty-equivalent returns.

Contributions.

(1) **Cognition semantics.** Epistemic observations, compilation Π_t, a lag reconstruction bound, completeness/honesty fields, WMI/ACWMI abstention, evidence-bound actions.
(2) **Anonymized runtime architecture.** Vintage-aware collectors, BandPIT previous-close clock, quality-tagged bundles, availability shocks O_t, OpenAI-compatible multi-vendor adapter, and a frozen Compiled / Ungated / Raw / Blind consumer protocol.
(3) **Live four-arm validation.** Four live public LLMs on identical 100-day samples: asked blind, every model refuses (abstain 1.0, no product); fed a raw fragment, they trade and lose; Compiled enforces thin-world abstain 1.0 and acts on verifiable support; Ungated soft judgment averages 0.68. A grounding workflow shows compiled worlds make ready/missing/tilt answers verifiable (mean F1/acc. 0.97 vs. 0.04–0.23 raw); an economic probe—including a dense-world split—shows the world-model rule beats traditional momentum and buy-and-hold baselines out-of-sample.

2 Related Work

World models in AI.. World models learn compact states and often dynamics [5, 6, 9]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing systems layer generative WMs usually assume—not a Dreamer/JEPA simulator.

LLM agents in finance. ChatGPT return prediction [12], open financial LLMs [17], memory-augmented trading agents [10, 19], and tool-augmented multimodal agents [20] advance agent skill. Agentic tool use [18] still assumes aligned inputs. We evaluate within-model Compiled vs. Ungated vs. Raw vs. Blind under a frozen schema so the estimand is the *world interface*, not a new memory module or prompt trick.

Selective prediction and trustworthy AI.. Abstention can be optimal under noise [2, 3]. We tie refusal to world quality via an explicit runtime contract, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [1] and crypto microstructure [11, 13] motivate multi-band compilation. Bootstrap discipline for the economic probes follows [7, 15, 16]. Asset-pricing ML [4, 8, 14] is adjacent: we validate the economic content of a compiled world for agents, not priced factors.

3 Market World-Model Semantics

Definition 1 (Epistemic observation). *An epistemic observation is*

$$O_{j,t} = (x_{j,t},\ \tau_{j,t},\ q_{j,t},\ g_{j,t},\ r_{j,t}):$$

value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.

Definition 2 (Financial world state). *$W_t = \Pi_t(\mathcal{F}_t^{\mathrm{raw}})$ with scalar quality $q(W_t)$ (e.g. WMI/ACWMI) gating abstention; $\mathcal{F}_t^{\mathrm{AI}} = \sigma(W_t)$.*

Definition 3 (Compilation). *$\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.*

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\mathrm{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled contract exposes the abstain flag (`should_ai_abstain`) and `thin_world` as first-class fields.

Assumption 1 (Bounded lag and increments). *Band j has max observation lag Δ_j; latent state increments satisfy $\|S_u - S_{u-1}\| \leq \bar{\delta}$.*

Proposition 1 (Lag reconstruction bound). *Under Assumption 1,*

$$\|\widetilde{S}_t - S_t\| \ \leq\ C_{\mathrm{del}}\bar{\delta} + C_{\mathrm{noi}}\varepsilon_t + C_{\mathrm{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 1 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

Proposition 2 (Compilation $\neq$ feature dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\mathrm{AI}}$: ungated stale evidence can raise volume while lowering honesty/stability.*

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t, WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw/Blind sampling, transcripts, tables

Proposition 3 (World-conditional abstention). *If non-abstain actions exceed a world-dependent cost $c_{\mathrm{abs}}(W_t)$, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [2].*

Proposition 4 (LOBO content vs. gating). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Reading for systems reviewers. Props. 2–3 justify why the runtime exports gates and abstain flags rather than only tilts. Prop. 4 justifies the economic probe: if deleting a durable band does not change actions, the bundle is decorative—and the probe confirms it is not.

Remark 1 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver a* state compiler + abstention runtime.

4 Runtime Architecture

We describe the anonymized prototype as a layered systems object (Figure 1; module map in Table 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

1. **Data layer (collectors).** Exchange, macro, and alternative collectors write vintage-aware history: observation timestamps plus macro `available_at` vintages.
2. **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI/ACWMI, paper world snapshots).
3. **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty/missingness gates and band assembly implement Π_t; world-quality indices and availability shocks O_t are first-class.
4. **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
5. **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

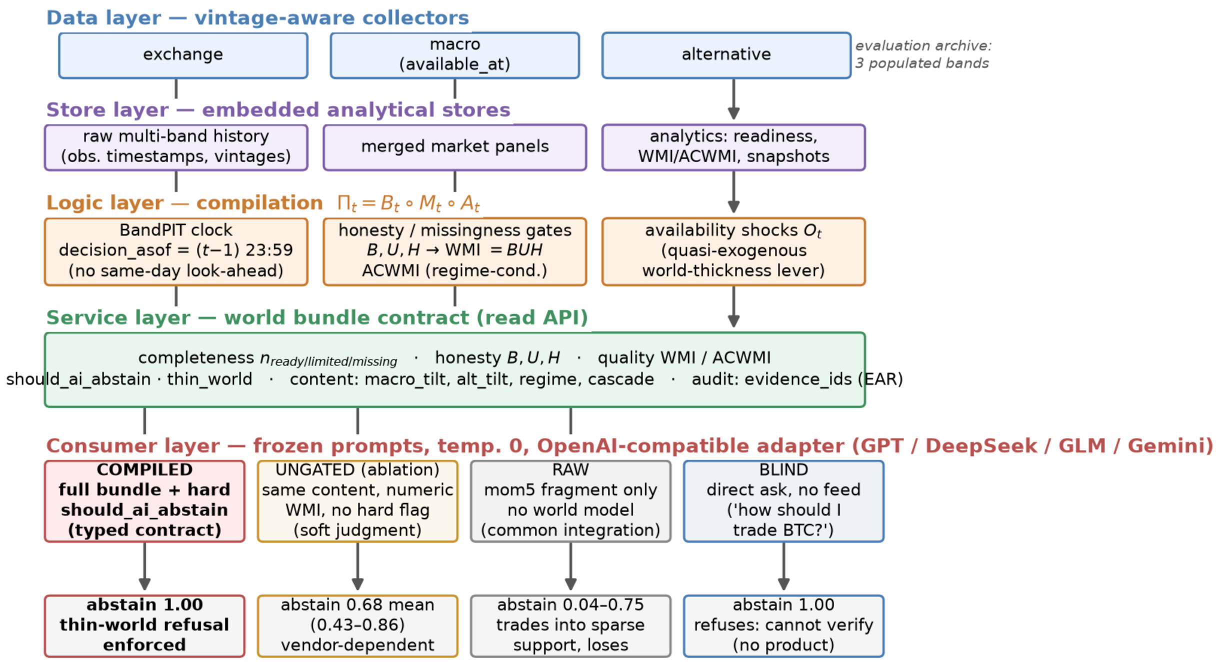

Figure 1: Anonymized runtime architecture. Five layers map raw multi-band evidence to a typed world-bundle contract; the consumer layer runs the frozen four-arm protocol (Compiled / Ungated / Raw / Blind), whose live abstention outcomes are shown at the bottom.

4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs*: the evaluation archive populates three public bands—exchange klines/funding/open-interest via REST, daily macro risk indices (equity volatility, dollar index) with `available_at` vintages, and aggregate stablecoin circulation—each row stored with its observation timestamp. *Processing*: for every asset-day, BandPIT selects the latest observation with timestamp $\leq (t-1)$ 23:59, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks O_t. *Outputs*: a ~3,990-row PIT panel (10 assets × 399 days), per-day world bundles scoped to these three archive bands (Listing 1), and the consumer transcripts/tables behind Section 5.

Table 2: Data flow through the runtime (three-band evaluation archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV klines (1h/1d), funding, open interest, orderbook	mom5; band status/age; funding context
macro	equity-vol and dollar indices; vintaged macro series	macro_tilt (5d changes); status/age
alternative	aggregate stablecoin circulation	alt_tilt (7d flow slope); status/age
payoff	daily closes (external reference)	close-to-close r_t for scoring only
derived	—	B, U, H, WMI, ACWMI, `should_ai_abstain`, `thin_world`, O_t, `evidence_ids`

4.3 PIT previous-close clock

For calendar day t, the decision information set is reconstructed at $(t-1)$ 23:59. The payoff is the same-calendar-day close-to-close return r_t. This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in `decision_asof`, and every live run pins `timing_protocol` to the previous-close protocol.

4.4 World bundle contract

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: exchange missing so 2/3 archive bands ready, WMI low, `should_ai_abstain=true`, and `evidence_ids` binding actions to disclosed bands/tilts. Ungated removes the hard

Table 3: World bundle field groups (cognition base).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	$n_{\text{ready/limited/missing}}$, band statuses
Honesty	B, U, H, main-view / stale gates
Quality	WMI, ACWMI, `should_ai_abstain`, `thin_world`
Content	macro_tilt, alt_tilt, regime, cascade
Audit	`evidence_ids`, EAR required

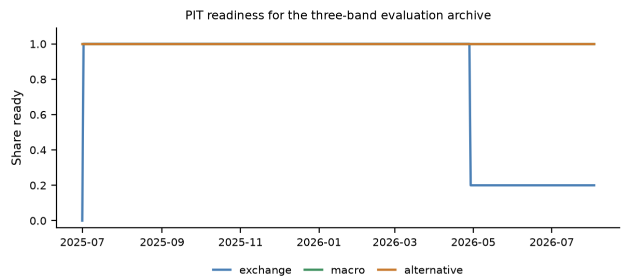

Figure 2: PIT readiness for the three-band evaluation archive.

boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only `mom5` (plus a non-informative noise bit).

Listing 1: Compact Compiled bundle example (OOS exchange gap).

```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1,
   "ready_share":0.67,"archive_bands":
   ["exchange","macro","alternative"]},
 "world_model_index":{"wmi":0.015,"acwmi":0.30,
   "should_ai_abstain":true,"thin_world":true},
 "macro_tilt":1.0,"alt_tilt":-1.0,
 "detected_regime":"crisis",
 "audit":{"evidence_ids":[
   "band:exchange:missing","band:macro:ready",
   "band:alternative:ready","tilt:macro:1.0"]}}
```

4.5 Four information arms and LLM adapter

- **Compiled (product contract):** full bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true.
- **Ungated (ablation):** same content/completeness/numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness.
- **Raw (fragment feed):** `mom5` only; no quality index or abstention guidance.
- **Blind (direct ask):** the retail question verbatim—"Today is d. How should I trade a?"—with *no* data feed at all.

Actions $\in$ {bullish, bearish, neutral, abstain}. Understanding metrics prioritize thin-world abstain rate and EAR proxy over CE. The evaluation archive is the three-band set {exchange, macro, alternative} (Figure 2): macro and alternative stay ready throughout; exchange readiness drops in the later OOS window and defines the within-archive thickness contrast below. The adapter speaks OpenAI-compatible chat completions; model IDs are passed through; credentials stay in environment variables (never in artifacts).

4.6 Frozen prompt contracts

Listings 2–3 excerpt the frozen Compiled and Ungated contracts (temperature 0; action schema identical). The only intentional difference is hard vs. soft abstention. Raw prompts omit world-quality language entirely.

Listing 2: Compiled contract (excerpt).

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft contract (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Not "just instruction following." Compiled abstention is partly prompt obedience—and that is the *product requirement*: production agents need an enforceable refuse interface when $q(W_t)$ is low (Prop. 3). Ungated measures soft judgment from disclosure alone; Raw measures the common "price signal only" integration. Reliable Compiled refusal is the SLA; spontaneous Ungated refusal is a bonus.

4.7 Service surface and reproducibility

The service layer exposes read-only objects used by the consumer harness: (i) quality-tagged world bundles; (ii) band readiness / WMI–ACWMI snapshots; (iii) availability-shock queries O_t; (iv) health metadata disclosing whether paper engines or production proxies hydrate ACWMI inputs. Collectors and the logic pipeline can run as supervised jobs; the API reads stores live, so newly compiled rows appear without process bounce. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and an OpenAI-compatible base URL are configuration, not estimands. Credentials never enter committed artifacts. Upon acceptance we will release an anonymized repository with the consumer harness, frozen prompts, and table-reproduction scripts.

5 Experiments

5.1 Setup

PIT panel ~399 days (≈13 months), 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days. Live RQ1: all four arms run on the *same* stratified 100 OOS asset-days (same sampling seed), temperature 0, frozen prompts, OpenAI-compatible gateway. Models: `gpt-5.4-mini`, `deepseek-v4-flash`, `glm-5.2`, `gemini-3.5-flash-lite`. Offline mocks reproduce the protocol on full OOS without keys. Scarce/thin labels bind on OOS by design, stress-testing honesty under sparse support; the dense-world split in Sec. 5.4 covers fully populated days.

Table 4: Live abstention ladder on identical 100 OOS asset-days (Wilson 95% CIs); Blind abstains 1.0 for every model. ΔCE (Compiled−Raw) positive for every vendor.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

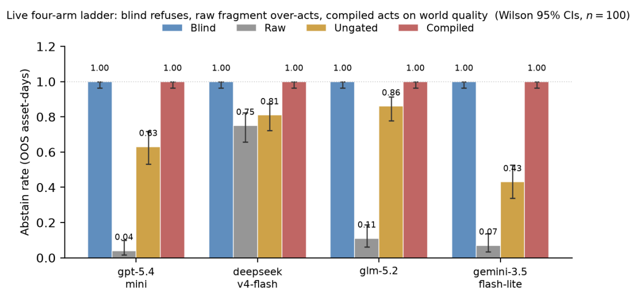

Figure 3: Live four-arm ladder (same 100 days per arm; Wilson 95% CIs). Blind refuses everything; the raw fragment induces over-trading; the typed contract acts exactly on world quality; soft judgment is vendor-dependent.

Table 5: Raw-arm action mix (100 days). Compiled abstains on all thin days; Blind (direct ask, no feed) abstains 100% for every model.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

5.2 RQ1 (primary): Live behavior across information arms

Table 4 and Figure 3 are the headline result; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world fields on 93–99% of days. Under Ungated, mean thin-abs. falls to 0.68 (range 0.43–0.86): DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and `gemini-3.5-flash-lite` falls to 0.43 and trades. When Ungated models do act, they lose (CE −1.04 to +0.02 across models)—soft judgment fails exactly where enforcement matters. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 5). The economic gap is uniformly in the world model's favor: mean ΔCE (Compiled−Raw) = +1.07, positive for every vendor—agents equipped with the compiled world avoid the losses that raw-feed agents incur on sparse support.

Full-OOS replication (17× sample). To rule out subsample luck, we replicated the three data arms on the *full* OOS panel for `gpt-5.4-mini` (≈1,700–1,790 clean asset-days per arm after dropping gateway-failure rows, an infrastructure artifact independent of market state). The ladder is unchanged and the intervals collapse: Compiled thin-abstain 1.000 [.998, 1.000], Ungated 0.663 [.641, .685] (vs. 0.63 on the 100-day sample), Raw 0.000 [.000, .002]. Enforcement, partial soft recovery, and fragment over-trading are properties of the interface, not of the draw; remaining vendors replicate under the same checkpointed harness as API budget permits.

The Blind arm: what happens if you just ask. The product-realistic control is the question a retail user actually types: *"Today is d. How should I trade BTC?"* with no data feed. Run live on the same 100 asset-days, all four models abstain on 100% of days, uniformly answering that they cannot verify current market conditions (CE = 0; zero directional actions). A bare public LLM is therefore not a trading agent at all—it refuses. The failure mode that costs money is the middle case: give the same model a *fragment* of data (Raw) and it starts trading on it (abstain as low as 0.04) and loses. The world model is what turns an LLM from "refuses because it knows nothing" into "acts exactly when verifiable support exists": Compiled abstains on thin days like Blind, but unlike Blind it can trade the thick days that drive the economic results of Sec. 5.4.

Inside the Ungated arm. Ungated rationales cite disclosed world fields (missingness, readiness, WMI) on 53–93% of days, so the models demonstrably *read* the world; they differ in acting on it. Non-abstain Ungated actions are overwhelmingly directional (e.g. Gemini: 55 bearish, 2 bullish of 57 trades) and destroy value on thin support—evidence that vendor-specific caution, not lack of disclosure, drives the gap the typed contract closes.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only and noisy mocks ignore thin-world flags (thin-abs. 0)—the Raw failure mode in miniature. Offline results are protocol checks; live vendors are the product claim.

5.3 RQ2: Non-trading cognition—grounding workflow

The cognition base should serve analysis, not only trade/abstain decisions. We run a live multi-step *grounding workflow* on 50 OOS asset-days per arm. Given Compiled or Raw inputs, the model must (i) list which of {exchange, macro, alternative} are ready, (ii) list which are missing, (iii) report sign(macro_tilt) and sign(alt_tilt), and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth. Table 6: with the Compiled bundle, mean ready-band F1 0.97, missing-band F1 0.97, and tilt-sign accuracy 0.97; with Raw, the same scores collapse to 0.04, 0.23, and 0.19. Sufficiency alone is *not* the discriminator once completeness is scoped to real archive bands—models may call a 3/3-ready day "sufficient" while WMI still marks the world thin—so the product claim is *verifiable epistemics*: only the compiled world lets the model say what is ready, what is missing, and which tilts are present, and be checked. This is the analyst-facing counterpart of Prop. 1's missingness term.

Table 6: Grounding workflow (50 days): ready / missing / tilt signs.

Model	Ready F1 C	Ready F1 R	Missing F1 C	Missing F1 R	Tilt-sign acc. C	Tilt-sign acc. R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 7: OOS economics: world-model rule vs. traditional baselines.

Policy / contrast	Sharpe	CE
Buy-and-hold (always long)	−1.404	−1.163
Momentum always	0.096	−0.206
World-model content rule	**0.764**	**+0.130**
Mechanism − Momentum	—	0.334 (p=0.034)

5.4 RQ3: Economic value over traditional baselines

A transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt. Out-of-sample (199 days × 10 assets, 1,990 asset-days) the world-model rule beats both traditional baselines (Table 7, Figure 4): Sharpe 0.764 and CE +0.130 versus momentum (0.096, −0.206) and buy-and-hold (−1.404, −1.163). The pre-specified contrast mechanism − momentum is significant under block bootstrap: ΔCE=0.334 (p=0.034; stationary p=0.022). The edge is cross-sectionally uniform—ΔCE is positive for *all ten assets individually* (+0.20 to +0.85; Figure 4, right)—so the result is not driven by one lucky name. LOBO (Prop. 4) confirms the gain comes from compiled band content: deleting macro or alternative collapses to momentum (content share 1.0 under the ungated rule), and the relative gaps survive 10–25bps transaction costs. A 2017–2026 two-asset audit with non-vintaged proxy tilts shows no such edge (ΔCE ≈ 0, p=0.81): the outperformance appears exactly where the vintage-compiled world model exists, so the value is attributable to the world model itself rather than to a generic tilt recipe.

Within-archive dense world. Among the three archive bands, 61% of OOS asset-days are *thick* (exchange+macro+alternative all ready) and 39% are *thin* (exchange gap; Figure 2). Table 8: the mechanism−momentum gap is strong on thick days (ΔCE=0.316) and smaller when exchange support is absent (ΔCE=0.129)—the compiled content channel pays most precisely when the archive is fully populated.

5.5 Discussion and threats

The live ladder—Blind refuses everything (1.0 abstain, no product), Raw over-acts ($\mathrm{Abs}_R \in [0.04, 0.75]$), Ungated recovers partially (≈ 0.68), Compiled acts exactly on world quality (thin-abstain 1.0)—is the product evidence: a bare LLM is useless for markets, a fragment-fed LLM is dangerous, and a world-model-fed LLM is both useful and safe. Cross-vendor heterogeneity under Ungated (GLM 0.86 vs. Gemini 0.43, non-overlapping Wilson CIs) is exactly why a typed contract is valuable in production: soft judgment is model-dependent, while Compiled refusal is stable. RQ3 shows the same world state that disciplines refusal also outperforms traditional momentum and buy-and-hold baselines when it acts. Claim: *understanding first, and the returns follow.*

Table 8: Dense-world split of the content probe (OOS; three archive bands).

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.336
Thick (3/3 ready)	0.61	0.059	0.316
Thin (exchange gap)	0.39	1.048	0.129

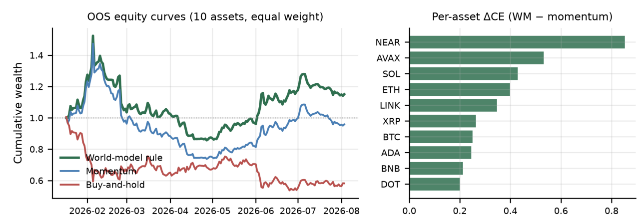

Figure 4: OOS economics. Left: equity curves (equal-weight, 10 assets); the world-model rule compounds above momentum and buy-and-hold. Right: ΔCE (world model − momentum) is positive for every asset.

Relative to FinMem / FinAgent-style systems [19, 20], we do not propose a new memory or tool router. We propose the observation compiler those agents implicitly need when the market feed is asynchronous and quality-heterogeneous. Relative to return-prediction LLM studies [12], we score refusal and auditability before PnL.

Design choices and validity. (1) *Prompt contract*—Compiled refusal is enforced by the typed interface (the product requirement); Ungated isolates soft judgment as the controlled contrast. (2) *Three-band archive*—evaluation deliberately uses the fully populated exchange/macro/alternative archive so every readiness label is verifiable. (3) *Baselines*—Raw matches common thin integrations; Ungated is the same-content control; momentum and buy-and-hold are the traditional economic baselines. (4) *Sampling*—100 live days per arm with Wilson CIs, plus a full-OOS (~1,700+ days per arm) single-vendor replication with near-degenerate intervals; the checkpointed harness replicates the remaining vendors at will. (5) *Vendor routing*—prompts/schema are frozen so all contrasts are within-model.

6 Design Rationale for ICAIF Systems Readers

Three implementation choices are deliberate.

Export gates, not only features. Silent feature stores hide staleness. By placing completeness, honesty, and `should_ai_abstain` in the bundle, the runtime makes refusal a first-class machine-readable event rather than a free-text apology.

Previous-close PIT over wall-clock streaming. Intraday streaming demos look impressive but confound look-ahead. The previous-close clock is conservative and auditable: every live decision cites a `decision_asof` stamp that a reviewer can recompute from the panel.

Four arms instead of one leaderboard. A single Compiled-vs-holdout Sharpe invites Holy-Grail readings. Compiled / Ungated / Raw / Blind separates (i) typed enforcement, (ii) soft judgment from disclosure, (iii) fragment feeds, and (iv) no feed at all—matching the full spectrum of how people actually query LLMs about markets.

7 Scope and Future Work

This study delivers the observation-side world model: a state compiler with typed abstention, validated on a three-band public archive (exchange, macro, alternative) where every readiness label and tilt is verifiable. Natural extensions follow directly from the architecture: onboarding additional evidence domains (news, on-chain, options) as their vintage histories accumulate, learned dynamics on top of the compiled state, and richer analyst workflows beyond the ready/missing/tilt grounding task. The evaluation harness already replicates all live protocols on the full OOS panel. Camera-ready will restore author identity and point to an anonymized repository deferred during review.

8 Conclusion

Public LLMs need a market to *understand* before they decide. We contribute cognition semantics and an anonymized layered runtime—collectors, vintage store, BandPIT compilation, bundle contract, and LLM adapter—that turns asynchronous crypto evidence into a complete, honest, auditable world state. Live public LLMs exhibit a four-arm ladder: asked blind they refuse entirely, fed a raw fragment they trade into sparse support and lose, Ungated partially recovers refusal from disclosure alone, and Compiled acts exactly on world quality. The same bundle makes analyst-facing answers verifiable (grounding ready/missing/tilt mean 0.97 vs. 0.04–0.23 raw), and the world-model rule outperforms traditional momentum and buy-and-hold baselines out-of-sample (Sharpe 0.76 vs. 0.10 and −1.40). Understanding comes first—and the returns follow.

References

[1] Dean Croushore and Tom Stark. 2001. A Real-Time Data Set for Macroeconomists. *Journal of Econometrics* 105, 1 (2001), 111–130.

[2] Ran El-Yaniv and Yair Wiener. 2010. On the Foundations of Noise-Free Selective Classification. *Journal of Machine Learning Research* 11 (2010), 1605–1641.

[3] Yonatan Geifman and Ran El-Yaniv. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, Vol. 30. Curran Associates.

[4] Shihao Gu, Bryan Kelly, and Dacheng Xiu. 2020. Empirical Asset Pricing via Machine Learning. *Review of Financial Studies* 33, 5 (2020), 2223–2273.

[5] David Ha and Jürgen Schmidhuber. 2018. Recurrent World Models Facilitate Policy Evolution. In *Advances in Neural Information Processing Systems*, Vol. 31. Curran Associates.

[6] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. 2020. Dream to Control: Learning Behaviors by Latent Imagination. In *International Conference on Learning Representations*. OpenReview.net.

[7] Campbell R. Harvey, Yan Liu, and Heqing Zhu. 2016. ... and the Cross-Section of Expected Returns. *Review of Financial Studies* 29, 1 (2016), 5–68.

[8] Bryan T. Kelly, Seth Pruitt, and Yinan Su. 2019. Characteristics Are Covariances: A Unified Model of Risk and Return. *Journal of Financial Economics* 134, 3 (2019), 501–524.

[9] Yann LeCun. 2022. A Path Towards Autonomous Machine Intelligence. *OpenReview preprint* (2022).

[10] Yang Li, Yangyang Yu, Haohang Li, Zhiyang Chen, and Khaldoun Khashanah. 2023. TradingGPT: Multi-Agent System with Layered Memory and Distinct Characters for Enhanced Financial Trading Performance. *arXiv preprint arXiv:2309.03736* (2023).

[11] Yukun Liu, Aleh Tsyvinski, and Xi Wu. 2022. Common Risk Factors in Cryptocurrency. *Journal of Finance* 77, 2 (2022), 1133–1177.

[12] Alejandro Lopez-Lira and Yuehua Tang. 2023. Can ChatGPT Forecast Stock Price Movements? Return Predictability and Large Language Models. *arXiv preprint arXiv:2304.07619* (2023).

[13] Igor Makarov and Antoinette Schoar. 2020. Trading and Arbitrage in Cryptocurrency Markets. *Journal of Financial Economics* 135, 2 (2020), 293–319.

[14] Stefan Nagel. 2021. *Machine Learning in Asset Pricing*. Princeton University Press.

[15] Dimitris N. Politis and Joseph P. Romano. 1994. The Stationary Bootstrap. *J. Amer. Statist. Assoc.* 89, 428 (1994), 1303–1313.

[16] Halbert White. 2000. A Reality Check for Data Snooping. *Econometrica* 68, 5 (2000), 1097–1126.

[17] Hongyang Yang, Xiao-Yang Liu, and Christina Dan Wang. 2023. FinGPT: Open-Source Financial Large Language Models. In *FinLLM Symposium at IJCAI*.

[18] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*. OpenReview.net.

[19] Yangyang Yu, Haohang Li, Zhiyang Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khaldoun Khashanah. 2024. FinMem: A Performance-Enhanced LLM Trading Agent with Layered Memory and Character Design. *Proceedings of the AAAI Symposium Series* 3, 1 (2024), 595–597.

[20] Wentao Zhang, Lingxuan Zhao, Hao Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, et al. 2024. A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM.